One Amos bound, three consumer sites: machine-checked Bessel-ratio calculus for lattice gauge expansions

Lluis Eriksson*

July 2026

Abstract

The Amos-type upper bound on the modified-Bessel ratio, $I_{\nu+1}(x)/I_\nu(x) < x/(\nu + \frac{1}{2} + \sqrt{(\nu+\frac{1}{2})^2 + x^2})$, enters lattice-gauge-theory expansions repeatedly: it drives a φ-monotonicity lemma in a two-dimensional surface expansion, a Feynman–Hellmann unit-step inequality for $(\log I_\nu)'$, and the exactly soluble string-tension lane. In our programme it was carried at three sites as textual copies of a named hypothesis, machine-checked against two different toolchains. This paper closes that fragmentation: a single Lean 4 module in the pinned mother repository defines the bound once (`AmosBound`) and proves, through that one definition, a two-lemma calibration engine and four consumer theorems — the φ unit step and φ-monotonicity step of the surface track, and the unit-step and log-derivative inequalities of the Feynman–Hellmann track — with rational non-vacuity witnesses whose Amos hypothesis holds by exact Pythagorean arithmetic. All eight statements carry the axiom oracle [`propext`, `Classical.choice`, `Quot.sound`]. A certified companion (interval arithmetic, 256-bit precision, self-contained series-plus-tail enclosures, committed transcript with dual-hash witness) verifies the bound provably strictly at all 1206 points of a pre-registered consumer-driven grid. The Amos bound itself remains a classical cited theorem: what is discharged here is the fragmentation — three scattered, differently-pinned copies become one named proposition with one oracle and one certified numerical witness — not the bound, and no downstream result changes its verification class.

1 Introduction

Cluster and character expansions of two-dimensional lattice gauge theory run on ratios of modified Bessel functions: the exact mean plaquette is $I_2(\beta)/I_1(\beta)$, the string tension is $-\log(I_2/I_1)$, and monotonicity structures of expansions in the heat-kernel/von Mises family reduce to inequalities between consecutive ratios $\rho_\nu = I_{\nu+1}/I_\nu$. The workhorse inequality is the Amos-type upper bound [1, 2, 3]

$$\rho_\nu(x) \;<\; \mathrm{A}(\nu, x) \;:=\; \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu+\frac{1}{2})^2 + x^2}} \qquad (x > 0,\ \nu \geq 0). \tag{1}$$

*This manuscript and parts of the formal development were produced with AI assistance under the repository's audit protocol. Every theorem below is machine-checked in Lean 4 against a pinned Mathlib; the grid certification of Section 5 is enclosed by committed interval arithmetic; remaining numerics are labelled paper-level where they occur. Repository: https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME.

In this programme, (1) entered three developments as a *named hypothesis* — an explicit assumption of a machine-checked theorem, never silently used: (a) the surface track's φ-monotonicity lemma, whose Lean core carried `hAmos` verbatim and on which the surface theorem's $E' < 0$ for $0 < \beta \leq 3.5$ is conditional; (b) the Feynman–Hellmann unit-step development for $(\log I_\nu)'$ [5], whose Lean core carried `hamos` verbatim; (c) the exactly soluble string-tension lane, whose certified numerical twin computes precisely the ratios of (1). Copies (a) and (b) were standalone files verified against Lean 4.30.0-rc2 with a Mathlib snapshot *different* from the mother repository's pin — statements shared by textual identity, not by a common definition, with no single artifact certifying both.

This paper contributes:

1. a single Lean module `AmosClosure` in the pinned mother core defining (1) once (Definition 2.1) and proving the calibration engine (Lemmas 3.1, 3.2) and all four consumer theorems (Theorems 3.3–3.6) through that one definition — the toolchain-fragmentation gap closed, with a clean axiom oracle on all eight statements;

2. rational non-vacuity witnesses (Section 6) in which the Amos hypothesis holds by *exact* arithmetic (a Pythagorean calibration point makes the square root rational);

3. a certified companion (Section 5) verifying (1) provably strictly at all 1206 points of a pre-registered grid covering the domain the consumers actually consume.

Statuses are kept separate throughout: **exact** means a Lean theorem in the pinned mother core with axiom oracle [`propext`, `Classical.choice`, `Quot.sound`]; **certified** means a committed interval-arithmetic transcript with ball-plus-boolean output; **paper-level** means ordinary prose, always flagged. The formal development lives in a repository whose top-level namespace (`YangMills/`) reflects its original motivation in lattice gauge theory; no reduction to any continuum or gauge-theoretic problem is claimed or used, and — stated plainly — *no downstream theorem changes its verification class by this paper*: consumers of (1) were conditional on a classical bound before and remain so; what improves is that the condition is now one audited object instead of three copies.

2 Setting

Throughout, I_ν is the modified Bessel function of the first kind; for $x > 0$ and $\nu \geq 0$ it is positive, and consecutive orders satisfy the three-term recurrence $I_{\nu-1}(x) - I_{\nu+1}(x) = \frac{2\nu}{x} I_\nu(x)$ [4, 10.29.1]. Write $\rho_\nu = I_{\nu+1}/I_\nu$. These facts are classical background (paper-level here); the Lean theorems below take the recurrence and the positivity of the relevant values as explicit hypotheses and are, as formal objects, statements of ordered-field algebra plus one square root.

Definition 2.1 (the bound, defined once)**.** *For $\nu, x, r \in \mathbb{R}$,*

$$\mathrm{A}(\nu, x) = \frac{x}{\nu + \frac{1}{2} + \sqrt{(\nu + \frac{1}{2})^2 + x^2}} \quad (\textit{Lean: } \texttt{amosRHS}), \qquad \texttt{AmosBound}\,\nu\;x\;r \;:\Longleftrightarrow\; r < \mathrm{A}(\nu, x).$$

Every consumer statement below hypothesizes `AmosBound` — never a textual copy of the right-hand side. The truth of `AmosBound` ν x $\rho_\nu(x)$ for actual Bessel ratios is the classical theorem of Amos [1] (sharp forms: [2, 3]); it is not *formalized here and remains the single analytic input of every consumer.*

3 Main results

All statements are in module `AmosClosure` (file `AmosClosure/Core.lean`) of the mother repository, compiled against its pinned toolchain; the recorded build and oracle facts are in Section 9.

Lemma 3.1 (calibration engine; `amos_calibration`). *Let $a, b > 0$ and $0 < t < b/(a+\sqrt{a^2+b^2})$. Then*

$$\frac{2a}{b} < \frac{1}{t} - t.$$

Lemma 3.2 (Amos smallness; `amos_small`). *If $\nu \geq 0$, $x > 0$, $r > 0$ and `AmosBound` $\nu\ x\ r$, then $r < x/(2\nu+1)$.*

Theorem 3.3 (φ unit step; `phi_unit_step`). *Let $m \geq 1$, $b > 0$, $u > 0$ with `AmosBound` $m\ b\ u$. Then*

$$\Big(\frac{1}{u^2} - u^2\Big) - \frac{2}{b}\Big((m-1)u + \frac{m+2}{u}\Big) + \frac{4(2m+1)}{b^2} > 0.$$

Theorem 3.4 (φ-monotonicity step; `phi_step_of_recurrences`). *Let $m \geq 1$, $b > 0$, and let $R_0 > 0$, $R_1 > 0$, R_2 satisfy the two recurrence relations $1/R_0 = R_1 + 2m/b$ and $R_2 = 1/R_1 - 2(m+1)/b$ (the shapes taken by $\rho_{m-1}, \rho_m, \rho_{m+1}$ under [4, 10.29.1]) together with `AmosBound` $m\ b\ R_1$. Then*

$$\varphi_m < \varphi_{m+1}, \qquad \text{where} \qquad \varphi_m = \frac{(m-1)/R_0^2 + (m+1)R_1^2}{m}.$$

Theorem 3.5 (unit step; `unit_step_of_recurrence_and_amos`). *Let $x > 0$, $\nu \geq 0$, $I_0, I_1 > 0$ and I_2 satisfy $I_0 - I_2 = \frac{2(\nu+1)}{x} I_1$ and `AmosBound` $\nu\ x\ (I_1/I_0)$. Then*

$$\frac{I_1}{I_0} - \frac{I_2}{I_1} < \frac{1}{x}$$

(positivity of I_2 is not needed).

Theorem 3.6 (log-derivative unit step; `logderiv_unit_step_increase`). *Under the hypotheses of Theorem 3.5, in the closed form $(\log I_\nu)' = \rho_\nu + \nu/x$,*

$$\Big(\frac{I_2}{I_1} + \frac{\nu+1}{x}\Big) - \Big(\frac{I_1}{I_0} + \frac{\nu}{x}\Big) > 0.$$

4 Proof architecture

The formal proofs are the authority; the mechanism is short enough to give almost in full.

One identity drives everything. Let $s = \sqrt{a^2+b^2}$ and $U = b/(a+s)$. Rationalizing, $U = (s-a)/b$ (from $(s-a)(s+a) = b^2$), and therefore

$$\frac{1}{U} - U = \frac{s+a}{b} - \frac{s-a}{b} = \frac{2a}{b}$$

exactly. Since $t \mapsto 1/t - t$ is strictly decreasing on $(0, \infty)$, any $t < U$ satisfies $1/t - t > 2a/b$: that is Lemma 3.1, and with $a = \nu + \frac{1}{2}$, $b = x$ it converts the Amos bound into the calibrated inequality $1/\rho - \rho > (2\nu+1)/x$. The unit step (Theorem 3.5) is then one line: the recurrence gives $I_0/I_1 - I_2/I_1 = 2(\nu+1)/x$ exactly, and subtracting the calibrated inequality leaves exactly $1/x$ of room. Theorem 3.6 is the same statement re-indexed.

The φ step. Writing u for the candidate ρ_m, the difference $\varphi_{m+1} - \varphi_m$ reduces, after substituting both recurrences and clearing denominators, to the exact algebraic identity

$$(m/R_1^2 + (m+2)R_2^2)\, m - ((m-1)/R_0^2 + (m+1)R_1^2)(m+1) \;=\; 2m(m+1)\,\delta,$$

$$\delta = \Big(\frac{1}{u^2} - u^2\Big) - \frac{2}{b}\Big((m-1)u + \frac{m+2}{u}\Big) + \frac{4(2m+1)}{b^2} = \Big(u + \frac{1}{u} - \frac{3}{b}\Big)\Big(\Big(\frac{1}{u} - u\Big) - \frac{2m+1}{b}\Big) + \frac{2m+1}{b^2}.$$

The second factor of the product is positive by the calibrated inequality; the first by the smallness $u < b/(2m+1) \le b/3$ (Lemma 3.2, itself one comparison of denominators); the additive term is positive outright. That is Theorems 3.3 and 3.4. Both exact identities are discharged in Lean by `field_simp; ring`; the inequalities by `nlinarith`/`linarith` over the ordered field. No analytic fact about I_ν enters anywhere: the analytic content is exactly the hypothesis `AmosBound` plus the recurrence shapes.

5 The certified companion

Script `scripts/c2_amos_arb.py` (SHA-256

`35a04b8c0f0532eafcee0d6559ebac330418d6bc29d6193e07bb36f32d1a39ce`

`python-flint` 0.9.0, precision 256 bits; transcript committed alongside it with a dual-hash witness, exit 0) certifies (1) for *true* integer-order Bessel ratios, pointwise and provably, on the grid pre-registered in the charter before the script existed:

$$\nu \in \{0, \dots, 20\} \times x \in \{0.25, 0.50, \dots, 14.00\} \text{ (1176 points)},$$
$$\nu \in \{0, \dots, 5\} \times x \in \{20, 50, 100, 200, 300\} \text{ (30 points)}.$$

The bulk box covers the surface-track consumption ($b = 4\beta c \le 14$ for $\beta \le 3.5$); the wide columns probe the large-x regime. Enclosures are self-contained: $I_n(x)$ is the truncated power series plus an explicit certified geometric tail majorant (term-ratio provably $< \frac{1}{2}$ at the truncation index, checked per point; all grid abscissae are dyadic, so the series starts from exact inputs), with outward rounding throughout and tri-state verdicts (PASS only when the strict inequality holds for every point of the enclosing balls; FAIL only when the reverse does; UNDECIDED otherwise).

Result (committed transcript): **all 1206 points pass; zero fail; zero undecided**. The measured slack floor — the smallest certified lower bound of $\mathrm{A}(\nu, x) - \rho_\nu(x)$ over the grid — is

$$2.782430724 \times 10^{-6} \pm 1.91 \times 10^{-16} \qquad \text{at } (\nu, x) = (0, 300),$$

consistent with the slack decreasing in x and with the programme's independent numerical twin (`exact2d.py`, outward-rounded decimal arithmetic), which reported a floor of 2.5×10^{-7} on its finer, wider box $[1, 100] \times [0.01, 300]$. Per the charter, this desk registered *no* prediction of the slack floor — the figure above is measured, and the twin-desk comparison is diagnostic-only. Scope: the certification is pointwise on the registered grid, for integer orders; it is a numerical witness that the classical hypothesis is comfortably true where the consumers consume it, not a proof of (1).

6 Non-vacuity, machine-checked

Both consumer families are instantiated in Lean at concrete rational values (file `NonVacuity.lean` of the module) — instance terms, not existentials. The calibration point is chosen Pythagorean: at order parameter $\frac{3}{2}$ and argument 2, $\sqrt{(\frac{3}{2})^2 + 2^2} = \sqrt{\frac{25}{4}} = \frac{5}{2}$ *exactly*, so $\mathrm{A} = \frac{1}{2}$ exactly and the Amos hypothesis for the witness ratio $\frac{2}{5}$ is the rational comparison $\frac{2}{5} < \frac{1}{2}$:

- φ family (`nonvacuous_phi_step`): $m = 1, b = 2, (R_0, R_1, R_2) = (\frac{5}{7}, \frac{2}{5}, \frac{1}{2})$; both recurrences hold by rational arithmetic, and the conclusion $\varphi_1 = \frac{8}{25} < \frac{7}{2} = \varphi_2$ is strict.
- unit-step family (`nonvacuous_unit_step`): $\nu = 1$, $x = 2$, $(I_0, I_1, I_2) = (1, \frac{2}{5}, \frac{1}{5})$; the recurrence is $1 - \frac{1}{5} = \frac{4}{5} = \frac{2\cdot 2}{2} \cdot \frac{2}{5}$, and the conclusion $\frac{2}{5} - \frac{1}{2} < \frac{1}{2}$ is strict.

What this proves: the hypothesis sets of all four consumer theorems are satisfiable with all-positive values and a strict Amos margin, so the theorems have non-trivial instances. What it does not prove: the witnesses are not true Bessel values (the certified companion is what ties the hypotheses to actual ratios), and satisfiability is not sharpness.

7 What the closure buys, site by site

Site	Before	After
Surface track φ-lemma (`surface-theorem/PhiMonotone.lean`)	standalone file, Lean 4.30.0-rc2 / Mathlib `cd3b69ba`, textual `hAmos`	statement re-proved in the mother pin through `AmosBound` (Theorem 3.4); original artifact untouched
Feynman–Hellmann core (`FHBessel/FHBesselAmos.lean`, [5])	standalone file, same off-pin toolchain, textual `hamos`, not wired into any lakefile	statements re-proved in the mother pin through `AmosBound` (Theorems 3.5, 3.6)
String-tension lane (`exact2d.py` twin)	certified ratio enclosures, no Amos check	the same ratios certified against (1) on the registered grid (Section 5)

The surface theorem's $E' < 0$ for $\beta \leq 3.5$ was conditional on the classical bound (1) before this paper and remains so; likewise the Feynman–Hellmann application [5] and the closure map of [7]. The gain is architectural and auditable: one definition, one oracle, one pin, one certified numerical witness — in place of three textual copies on two toolchains with none.

8 Related work

Bounds of the form (1) originate with Amos [1], who also gave the computational scheme for the ratios; Segura [2] derived sharp two-sided bounds and the associated Turán-type inequalities; Ruiz-Antolín and Segura [3] gave the sharp family at both ends of the parameter range. The recurrence is [4, 10.29.1]. On the programme side, the prose companions of the two Lean consumer families are the author's preprints [5] (whose "machine-checked core" is precisely the file ported here) and [7] (the surface-expansion closure map consuming the φ-lemma); a weighted Turán-type variant via the calibrated Amos bound is [6]. The formalization builds on Mathlib [8]. We are not aware of prior *formalized* treatments of Amos-type ratio bounds or their expansion-theoretic consumers; readers aware of one are invited to point us to it.

9 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

`07642720480157414db592fa85b626dafb71355b`

(lakefile and manifest agree); `python-flint` 0.9.0 on CPython 3.12.6; `tectonic` 0.15.0.

Recorded build facts (raw transcripts committed the day they were produced; the repository's `*.log` ignore rule is deliberately dodged by `.txt` naming): `lake build AmosClosure` completed with exit code 0 in **8161 jobs** at the Lean-integration commit

`b0aa4f7017e5aeb5a7e923fb13d9b0e8839632d9`

(release tag `c2-v1.0` on the final manuscript commit; the port from the off-pin sources compiled with zero API drift on the first run — raw transcript committed as

`scripts/c2_lake_build_run1_GREEN_transcript.txt`).

The axiom oracle (`lake env lean AmosClosure/Oracle.lean`; log committed as `c2_oracle_output.txt` in `scripts/` at the same commit) reports, for each of the eight theorems of Sections 3 and 6, exactly

`[propext, Classical.choice, Quot.sound]`.

Theorem-to-artifact map (all Lean files in module `AmosClosure`):

Result	Lean name	File	Status
Lem. 3.1	`amos_calibration`	`Core.lean`	exact
Lem. 3.2	`amos_small`	`Core.lean`	exact
Thm. 3.3	`phi_unit_step`	`Core.lean`	exact
Thm. 3.4	`phi_step_of_recurrences`	`Core.lean`	exact
Thm. 3.5	`unit_step_of_recurrence_and_amos`	`Core.lean`	exact
Thm. 3.6	`logderiv_unit_step_increase`	`Core.lean`	exact
§6 witnesses	`nonvacuous_*`	`NonVacuity.lean`	exact
§5 grid	—	`scripts/c2_amos_*`	certified
(1) for true I_ν	—	[1, 2, 3]	classical, cited
§2 Bessel background	—	[4]	paper-level

From a clean clone at the release revision:

```
lake exe cache get
lake build AmosClosure
lake env lean AmosClosure/Oracle.lean
python scripts/c2_amos_arb.py
tectonic -X compile papers/c2-amos-closure/c2_amos_closure.tex
```

Dual hashes (LF blob and CRLF worktree) for the companion transcript are committed as `scripts/c2_amos_transcript_HASHES.txt`; canonical artifact hashes for the release are in `papers/c2-amos-closure/RELEASE-MANIFEST.md`. Compare against `git show <rev>:path`, never a worktree checkout.

10 Conclusion

Eight machine-checked theorems route every programme consumer of the Amos bound through one audited definition in one pinned core, with rational satisfiability witnesses and a certified 1206-point numerical witness on the consumed domain. Limitations, stated where they arise: the bound (1) itself is classical and not formalized — it remains the single analytic input, and formalizing it (via a Mathlib theory of I_ν) is the natural next formal target; the grid certification is pointwise, not uniform; no downstream theorem changes its verification class. Reasonable next uses: consuming `AmosBound` directly in any future surface-track or transfer-matrix formalization, and upstreaming a modified-Bessel API to Mathlib.

References

[1] D. E. Amos, *Computation of modified Bessel functions and their ratios*, Math. Comp. **28** (1974), 239–251.

[2] J. Segura, *Bounds for ratios of modified Bessel functions and associated Turán-type inequalities*, J. Math. Anal. Appl. **374** (2011), 516–528. doi:10.1016/j.jmaa.2010.09.030.

[3] D. Ruiz-Antolín and J. Segura, *A new type of sharp bounds for ratios of modified Bessel functions*, J. Math. Anal. Appl. **443** (2016), 1232–1246. arXiv:1606.02008.

[4] NIST Digital Library of Mathematical Functions, https://dlmf.nist.gov/, Release 1.2.x; F. W. J. Olver et al., eds.

[5] L. Eriksson, *Recurrence–Amos proof of the unit-step order-monotonicity of* $(\log I_\nu)'$, *with a Feynman–Hellmann application to two-dimensional lattice gauge theory*, preprint, ai.viXra.org:2607.0020, July 2026.

[6] L. Eriksson, *A weighted Turán-type monotonicity lemma for modified Bessel functions via the calibrated Amos bound, with an application to the ordering of a surface expansion in two-dimensional lattice gauge theory*, preprint, ai.viXra.org:2607.0017, July 2026.

[7] L. Eriksson, *Ratio monotonicity for a killed von Mises bridge: exact bridge structure, certified negative results, a single-Bessel reduction, and a two-scale closure map for a surface expansion in two-dimensional lattice gauge theory*, preprint, ai.viXra.org:2607.0023, July 2026.

[8] The mathlib Community, *The Lean mathematical library*, in: Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020), ACM, 2020, pp. 367–381. doi:10.1145/3372885.3373824.